

Mathematics concept required for Deep Learning

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Why is Math required for Deep Learning?

Interested people who have the thirst to learn more about the concept behind a deep learning algorithm need to tackle Mathematics in some path of the way or another. Math is the core concept from which Deep Learning algorithms are built upon and is used to express the idea that seems quite obvious, but these are unexpectedly hard to elaborate and once it is elaborated properly, we can gain a proper understanding of the problem that we are given to solve. In this article, we are going to discuss in detail about the math required for Deep Learning. Now if there is a spark a light inside you, to learn more about deep learning then start with these Math topics:

1. Geometry and Linear Algebra

- Geometry of Vectors
- Angles and Dot Products with Cosine similarity
- Hyperplanes
- The geometry of Linear Transformation
- Rank of Matrix
- Linear Dependence
- Invertibility
- Determinant

Learn Linear Algebra [here](#) and basics on Geometry [here](#)

2. Matrix Decomposition

- Finding Eigenvalues and Eigenvectors
- Decomposing Matrices
- Operations on Eigendecomposition
- Single Value Decomposition
- Principal Component Analysis
- Matrix Approximation
- Eigendecomposition and Diagonalization of Symmetric Matrices
- Gershgorin circle theorem
- [LU Decomposition](#), QR Decomposition/Factorization
- Symmetric Matrices, Orthogonalization and Orthonormalization

Learn about Matrix Decomposition [here](#)

3. Single Variable Calculus

- Differential Calculus
- Rules of calculus

Learn calculus [here](#)

4. Probability and Distributions

- Sum Rule, Product Rule, and Bayes Theorem

- Gaussian Distribution
- Discrete and Continuous Probabilities
- Combinatorics
- Conditional and Joint Distributions
- Bernoulli Distribution
- Discrete Uniform Distribution
- Binomial Distribution
- Poisson Distribution
- Continuous Uniform Distribution
- Gaussian Distribution
- Exponential Family Distribution

Refer [this](#) for videos on Probability and for different distributions [here](#).

5. Naive Bayes

- Optical Characters Recognition
- Probabilistic Model for Classification
- Naive Bayes Classifier

To learn more about this refer to GeeksForGeek's article [here](#).

6. Multivariate Calculus

- Higher-dimensional Differentiation
- Multivariate chain rule
- Backpropagation Algorithm
- Gradient, Gradient Descent, and its geometry
- Integral Calculus, Partial Derivatives
- Vector-Values Functions
- Jacobian, Laplacian, Lagrangian Distribution

Refer [this](#) book to learn more about it

7. Integral Calculus

- Theorem of Calculus
- Sign Conventions
- Geometric interpretation
- Multiple integrals Concepts and Change of Variables

For basics of integral calculus refer [here](#).

8. Random Variables

- The concept of Discrete to Continuous in [Random Variables](#)
- Means, Variances, Standard Deviation and its concept on a continuum
- Probability Density Function
- Cumulative Distribution Function
- Joint Density Function
- Marginal Distribution
- Covariance and Correlation

An in-depth guide to this topic [here](#)

9. **Statistics**

- Evaluation and Comparison of Estimators
- Conducting Hypothesis Tests
- Constructing Confidence intervals

Refer [here](#) to learn more about it.

10. **Probability on Maximum Likelihood**

- Principle of Maximum Likelihood with examples
- Numerical optimization and Negative Log-Likelihood
- Maximum Likelihood for Continuous Variables
- Moment Generating Function
- Prior and Posterior, Maximum a Posteriori Estimation, Sampling Methods

To learn more about it refer